KAMALESH SURESH KUMAR

INTELLIGENT SYSTEMS & AUTOMATION ENGINEER

kamalsuresh2005@gmail.com | India | github.com/Kamalesh-Suresh-Kumar | Portfolio: kamalesh-suresh-kumar.github.io/Portfolio-KAI-HQ/

PROFESSIONAL SUMMARY

Computer Science undergraduate focusing on intelligent automation, deterministic intent resolution engines, and automated knowledge pipelines. Developer of client-side natural language routers capable of sub-10ms query-to-route resolution with zero external API dependencies. Deep foundational skills in algorithms, TypeScript, Python, and CI/CD automation, with an architectural emphasis on low latency, privacy-preserving static systems, and clean interfaces for generative model integration.

TECHNICAL SKILLS & CORE COMPETENCIES

Languages: TypeScript, Python, JavaScript, C/C++, SQL, Bash
Systems & Frameworks: React 19, Node.js, Vite, GitHub Actions CI/CD
Concepts: Intent Classification, Fuzzy Search, Sub-10ms Query Dispatching, State Mediators
Tooling: Git, Linux, JSON Schema, Modular Compilation

Key Competencies: Deterministic Intent Routing & NLP * Knowledge Engine Architecture & Indexing * Automated CI/CD Workflow Engineering * Data Structures & Complexity Optimization * Static Site Automation & Schema Validation * Modular Systems Architecture

KEY ENGINEERING PROJECTS

Rule-Based Conversational Router & Knowledge Dispatcher | *TypeScript, NLP Algorithms, Intent Routing, Deterministic Dispatch*
- Engineered a client-side intent classifier mapping natural language visitor prompts to physical facility coordinates and knowledge entities in under 10ms.
- Designed clean modular interfaces ensuring seamless drop-in extensibility for future Large Language Model (LLM) agents without restructuring core routing.

Automated CI/CD Knowledge & Build Pipeline | *GitHub Actions, Node.js, Vite, Bash, YAML*
- Built a continuous verification and build system validating TypeScript schemas and generating static production bundles on push events.
- Configured multi-environment deployment adaptations targeting GitHub Pages subpaths with zero downtime.

KAI HQ — Headquarter Operating System | *React 19, TypeScript, Tailwind CSS, Vite*
- Designed and shipped an 8-room portfolio operating system using a stateful Experience Engine and typed static Knowledge Engine.

EXPERIENCE & LEADERSHIP

Technical Trainee / Intern — Infosys SpringBoard *(2023 — 2024 | Virtual / India)*
- Mastered enterprise problem-solving paradigms, cloud computing infrastructure, and modular software engineering.

Software Development Intern — Viyuga Innovations *(2024 | India)*
- Implemented frontend and data-handling routines under strict engineering guidelines.

Event Team Member / Coordinator — IEEE Computer Society *(2023 — Present | India)*
- Organized technical workshops, student hackathons, and software engineering competitions.

EDUCATION

B.E. in Computer Science and Engineering — Undergraduate Engineering Program *(2022 — Present | India)*
- Core Coursework: Systems Architecture, Data Structures, Algorithms, Operating Systems, Computer Networks, Database Systems, Network Security.

CERTIFICATIONS & VERIFIED HONORS

- Enterprise Software & Cloud Computing Certification — Infosys SpringBoard (2023 — 2024)
- Li-Fi Penetration Testing Demonstration & Law Enforcement Briefing — Cybercrime Police Department (2024)
- IEEE Computer Society Event Team Contribution Award — IEEE Chapter (2023 — 2024)
- NCC Naval Wing Cadet Leadership & Service — National Cadet Corps (2022 — Present)